Conjecture 00000001294: three real branches every seven levels

lizaixi01, with OpenAI Codex assistance

23 October 2026

1 The actual inverse-tree convention

The claimed bound is false for the standard finite-depth unfolded inverse tree rooted at 1, for both normalizations

$$T_0(n) = \begin{cases} n/2 & n \text{ even}, \\ 3n+1 & n \text{ odd}, \end{cases} \qquad T_1(n) = \begin{cases} n/2 & n \text{ even}, \\ (3n+1)/2 & n \text{ odd}. \end{cases}$$

Applegate–Lagarias [1], §2, pp. 4–5, defines level N by the actual fiber $\{x \geq 1 : T_1^N(x) = a\}$ and explicitly permits cyclic-root labels to recur at different levels. A leaf here is a frontier node at depth N of this finite-depth tree. It is not a first-hit tree with root-cycle paths deleted. No conclusion about that different convention, or the Collatz conjecture, is needed.

Let $L_b(N) = \{x \geq 1 : T_b^N(x) = 1\}$ and $L_b^*(N) = \{x \in L_b(N) : 3 \nmid x\}$. Lean proves $x \leq 2T_b(x)$ and hence $x \leq 2^N T_b^N(x)$. Thus $L_b(N)$ is the complete actual fiber, represented as a filtered finite interval $[0, 2^N]$, not an assigned branching count. The pruned set is an actual subset. Every positive nonmultiple of 3 stays a positive nonmultiple of 3 under either forward map. Therefore all intermediate labels from any $x \in L_b^*(N)$ also survive the 3-pruning used in [1], equation (2.1).

2 Three distinct genuine preimages in seven steps

For every $n > 0$ with $3 \nmid n$, write $n = 9q + r$, where $r \in \{1, 2, 4, 5, 7, 8\}$. In the following table each listed pair (x, k) satisfies x odd and $3x + 1 = 2^k n$:

r	(x_1, k_1)	(x_2, k_2)
1	$(12q+1, 2)$	$(48q+5, 4)$
2	$(6q+1, 1)$	$(24q+5, 3)$
4	$(12q+5, 2)$	$(192q+85, 6)$
5	$(24q+13, 3)$	$(96q+53, 5)$
7	$(48q+37, 4)$	$(192q+149, 6)$
8	$(6q+5, 1)$	$(96q+85, 5)$

For T_0, take the three children $128n$, $2^{6-k_1}x_1$, $2^{6-k_2}x_2$. For T_1, take $128n$, $2^{7-k_1}x_1$, $2^{7-k_2}x_2$. Each is positive, not divisible by 3, and maps to n after exactly seven actual iterates. The three values are pairwise distinct. Lean verifies every residue case arithmetically and proves the iterate identities from $T_b(2x) = x$.

3 True cardinalities and the full asymptotic contradiction

Iterate this construction from the actual root 1. Every word $w \in \{0,1,2\}^m$ determines a genuine label $v_b(w) \in L_b^*(7m)$. This map is injective: if two final labels coincide, apply the actual map T_b^7 to recover their parents. Induction identifies their preceding words, and the three-child injectivity identifies the remaining symbol. In particular, children of distinct parents cannot collide. The formal finite type injection into the actual fiber gives

$$3^m \le |L_b^*(7m)| \le |L_b(7m)| \qquad (m \ge 0,\ b = 0,1).$$

Repeated labels at different levels of the unfolded tree cause no problem; injectivity is at each fixed level.

Put $c = \sqrt{4/3}$. The exact rational inequality

$$(c^7)^2 = (4/3)^7 < 9$$

gives $0 < c^7 < 3$. Consequently $r = c^7/3$ lies in $(0,1)$ and $Cr^m \longrightarrow 0$ for every real C. For every constant C and every cutoff N, choose $m \ge N$ with $Cr^m < 1$. Then $n = 7m \ge N$ satisfies

$$Cc^n < 3^m \le |L_b^*(n)| \le |L_b(n)|.$$

This refutes every eventual big-O bound, rather than merely checking finitely many levels or estimating a growth rate from data.

The final Lean statements prove the actual negations

$$|L_b^*(n)| \ne O(c^n), \qquad |L_b(n)| \ne O(c^n) \quad (b = 0,1)$$

using Mathlib's `Asymptotics.IsBigO` on the real-valued cardinality sequences and the actual filter `atTop`. The counterexample is for the standard unfolded finite-depth inverse tree; no claim that every positive integer eventually reaches 1 is assumed or proved.

4 Verification and disclosure

Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are pinned. The package supplies a new root build, transitive axiom audits of every theorem and a full official `leanchecker --fresh Main` replay. There is no `sorry`, `native_decide` or additional axiom. OpenAI Codex assisted; a separate supervising Codex agent reviewed the proof, source convention, report and PDF. This is AI review, not human certification. Organizer acceptance is pending.

References

[1] D. Applegate and J. C. Lagarias, *Density Bounds for the* $3x+1$ *Problem I. Tree-Search Method*, December 15, 1993, §2, pp. 4–5. `https://websites.umich.edu/~lagarias/doc/applegateI.pdf`. The shortcut map is equation (1.1); the unfolded cyclic-root and 3-pruned conventions are explicitly given in §2.